Design system and brand kit, v2.0

vgl: Virtual Globe Library

Visual identity, geometry, colour and typography guidelines

3DGEP / VGL
License: MIT (code)
Target: C99 ABI / C++20
Status: approved production kit
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01 Brand overview and positioning

vgl (Virtual Globe Library) is a lightweight, open-source C++ library with a stable C API for rendering a virtual globe inside an existing host application. It streams level-of-detail (LOD) terrain and imagery on the WGS84 ellipsoid, with backends for OpenGL 3.3, OpenGL ES 3.0, Direct3D 11, Ogre and WebGL 2 (Emscripten). The identity turns that architecture into a geometric, high-contrast visual language that stays legible from a 16×16 px browser tab up to high-DPI documentation banners.



Figure 1.1. Master lockup (`svg/vgl-banner.svg`): the globe-and-quadtree icon with the spurless vgl wordmark and justified subtitle.

02 Core design pillars

1. C-first and predictable `vgl_*`

Always lowercase **vgl**: in the logotype, CMake targets (`vgl::vgl`, `vgl::gl`), the C API prefix (`vgl_context_create`) and CLI tools (`vgl-gl`, `vgl-bake`). No decorative gradients, drop shadows or hairline wireframes.

2. Quadtree LOD on the WGS84 ellipsoid `F2 / F3`

A full 360° graticule paired with a 2×2 quadtree overlay in the upper-right quadrant: the tile pyramid the library streams, and the runtime debug views (`F2` wireframe, `F3` colours by tile level).

3. Zero-dependency portability `SVG 1.1`

Like the runtime, the assets need nothing extra. Every SVG outlines its text and uses only `rect`, `circle`, `ellipse` and `path` (knockout variants add one `clipPath`), so they render the same on every platform and browser.

4. Precision from orbit to ground `64×64`

The icon is drawn on a 64×64 master grid with 4-unit strokes and a dark keyline between globe and tiles. Below 32 px a separate pixel-fitted 16 px drawing takes over, so the favicon stays sharp in browser tabs.

03 Master logo and wordmark anatomy

The lockup is defined by one master file, `svg/vgl-banner.svg`, and the icon by `svg/vgl-icon.svg`. Every logo, wordmark and share image in the kit is generated from these two files. Never redraw, re-align or re-proportion the parts; edit the master and regenerate.



Figure 3.1. Construction of the master lockup in banner units (canvas 340.47 × 183.94). Amber rails mark outer ink edges; cyan circles show the g bowl (r 17) and hook (r 18).

Icon geometry (64×64 master) `1:1`

- **Tile:** 64×64, `rx 14`, Orbital Slate.
- **Sphere:** centre `(31.91, 32.09)`, `r 24.11`, Deep Ocean.
- **Graticule:** 4-unit Bathymetry Blue strokes: meridian ellipse `rx 11.35 ry 23.77`, prime meridian, equator, parallels at `y 20.2` and `y 43.97`.
- **Horizon ring:** `r 24.21`, 4-unit Atmosphere Cyan, unbroken 360°.
- **Quadtree keyline:** `x 32, y 4`, 28×28, `rx 4.71`, Orbital Slate.
- **Tiles:** four 10×10, `rx 2.5`, 2-unit gutter: NW cyan, NE amber, SW emerald, SE cyan.

Wordmark geometry (spurless classic) `11.5 stroke`

- **Strokes:** 11.5 units, round caps and joins. Dark mode Starlight, light mode Orbital Slate.
- **v:** two strokes meeting at a rounded apex; 38 × 34 on centrelines.
- **g:** closed circular bowl (`r 17`). The stem leaves the bowl tangentially at its **midline**, runs down 27 units and ends in an `r 18` hook. No top-right spur.
- **Dot:** concentric with the bowl, `r 6.2`; cyan on dark, Bathymetry Blue on light.
- **l:** 52-unit stem turning right into an `r 12` tail on the baseline.
- **Spacing:** 9 units v to g, 11 units g to l (outer ink).
- **Subtitle:** VIRTUAL GLOBE LIBRARY, Lato Bold caps, outlined, tracked so its ink runs exactly from the outer left of v to the outer right of l.

04 Logo variants, clear space and minimum size

Pick the variant by the surface it sits on, not by taste. Transparent files are cropped to the ink plus 8 units of padding; add the clear space below in your layout.



vgl-logo-dark.svg
Horizontal lockup for dark surfaces (GitHub dark, Orbital Slate, Deep Ocean).



vgl-logo-light.svg
Horizontal lockup for white and light-grey surfaces.



vgl-logo-stacked-dark.svg
Narrow spaces: docs sidebar, title pages.



vgl-wordmark-light.svg
Letters only, where the icon already appears nearby.



vgl-icon.svg
App and window icon; favicons, avatars.



Figure 4.1. Clear space: ½x on every side, where x is the outer height of the g bowl (45.5 banner units).

Variant	Minimum size
Horizontal lockup	200 px / 50 mm wide. Below that the subtitle drops under 5 px cap height; switch to the wordmark.
Stacked lockup	120 px / 30 mm wide
Wordmark	48 px / 12 mm wide
Icon	32 px from the master; 16 px only with `favicon-16.svg` or `favicon.ico`

Backgrounds: use the dark variants on Orbital Slate, Deep Ocean or any surface darker than #334155, and the light variants on white or light grey. Never place the lockup over photography or busy imagery without an Orbital Slate panel behind it.

04 Misuse

These are the most common ways the identity breaks. Each one is easy to avoid by using the generated files unchanged.



Don't stretch, squash or skew the lockup. Scale it uniformly.



Don't add a top-right spur to the g. The stem starts at the bowl's midline.



Don't recolour or reorder the quadtree tiles.



Don't add drop shadows, glows, bevels or gradients.



Don't put the dark-background logo on light surfaces. Use the light variant.



Don't resize or move the icon inside the lockup. Use the master files as-is.

Also avoid: retyping the subtitle as live text or in another typeface, rotating or cropping the globe, outlining the wordmark strokes, using the old v1.0 icon geometry, and placing anything inside the clear space.

05 Colour system and tokens

Every colour maps to a layer the library renders: orbital space, ocean bathymetry (GEBCO), atmospheric scattering, the digital elevation model (Copernicus DEM) and quadtree LOD telemetry (`F3` debug view).

Orbital Slate
#0B132B
RGB 11, 19, 43

Deep Ocean
#0F1D40
RGB 15, 29, 64

Bathymetry Blue
#0284C7
RGB 2, 132, 199

Atmosphere Cyan
#38BDF8
RGB 56, 189, 248

DEM Emerald
#10B981
RGB 16, 185, 129

Telemetry Amber
#F59E0B
RGB 245, 158, 11

Colour and token	HEX	RGB	CMYK (proof)	vs Orbital Slate	vs white	Assignment
Orbital Slate `--vgl-orbital`	`#0B132B`	11, 19, 43	90, 78, 42, 72	base	18.38:1 AAA	Icon tile, banner and social canvas, quadtree keyline, light-mode wordmark
Deep Ocean `--vgl-ocean`	`#0F1D40`	15, 29, 64	92, 76, 30, 52	1.11:1 Decorative only	16.53:1 AAA	Sphere fill behind the graticule, raised dark surfaces
Bathymetry Blue `--vgl-blue`	`#0284C7`	2, 132, 199	85, 35, 0, 0	4.49:1 AA large / UI	4.10:1 AA large / UI	Graticule (meridians, equator, parallels), light-mode g dot and links
Atmosphere Cyan `--vgl-cyan`	`#38BDF8`	56, 189, 248	66, 9, 0, 0	8.58:1 AAA	2.14:1 Decorative only	Horizon ring, NW and SE tiles, dark-mode g dot and links
DEM Emerald `--vgl-emerald`	`#10B981`	16, 185, 129	78, 0, 62, 0	7.25:1 AAA	2.54:1 Decorative only	SW tile (terrain), dark-mode success state
Telemetry Amber `--vgl-amber`	`#F59E0B`	245, 158, 11	0, 42, 96, 0	8.56:1 AAA	2.15:1 Decorative only	NE tile (finest LOD), dark-mode warnings and highlights
Starlight `--vgl-starlight`	`#F8FAFC`	248, 250, 252	2, 1, 0, 0	17.57:1 AAA	1.05:1 Decorative only	Dark-mode wordmark and headings
Geoid Slate `--vgl-geoid`	`#94A3B8`	148, 163, 184	proof needed	7.17:1 AAA	2.56:1 Decorative only	Dark-mode subtitle and secondary text
Slate 600 `--vgl-slate-600`	`#475569`	71, 85, 105	proof needed	2.43:1 Decorative only	7.58:1 AAA	Light-mode subtitle and secondary text (new in v2.0)

Contrast is WCAG 2.x relative luminance. AA body text needs 4.5:1; 3:1 is enough for large text (18 pt, or 14 pt bold) and UI graphics. Cyan, emerald and amber fail on white, so light surfaces use Bathymetry Blue, #047857 and #B45309 for links and states. CMYK values are carried over from v1.0 as starting points: proof them against your printer's profile.

05 Tokens in code

Use the primitive tokens for brand artwork and the semantic tokens for interfaces and documentation sites. The semantic set switches between light and dark automatically with `prefers-color-scheme`, and can be forced with `data-theme="light"` or `data-theme="dark"` on the root element.

Primitive tokens and type stacks

```
:root {
  --vgl-orbital: #0B132B;  --vgl-ocean: #0F1D40;
  --vgl-blue: #0284C7;     --vgl-cyan: #38BDF8;
  --vgl-emerald: #10B981;  --vgl-amber: #F59E0B;
  --vgl-starlight: #F8FAFC; --vgl-geoid: #94A3B8;
  --vgl-slate-600: #475569;
  --vgl-font-display: "Space Grotesk", "Inter", system-ui, sans-serif;
  --vgl-font-body: "Inter", system-ui, sans-serif;
  --vgl-font-mono: "JetBrains Mono", ui-monospace, Consolas, monospace;
}
/* semantic tokens switch on prefers-color-scheme and
   [data-theme]; full file: css/vgl-tokens.css */
```

Semantic tokens

Token	Light (default)	Dark
`--vgl-surface`	#FFFFFF	Orbital Slate
`--vgl-surface-raised`	#F1F5F9	Deep Ocean
`--vgl-text`	Orbital Slate	Starlight
`--vgl-text-muted`	Slate 600	Geoid Slate
`--vgl-link / --vgl-accent`	Bathymetry Blue	Atmosphere Cyan
`--vgl-success`	#047857	DEM Emerald
`--vgl-warning`	#B45309	Telemetry Amber
`--vgl-border`	#E2E8F0	#1E2A4A

The light-mode success (#047857) and warning (#B45309) values are functional extensions outside the brand palette: emerald and amber are too light for text on white. Full definitions: `css/vgl-tokens.css`.

06 Typography system

Display `Sans`

Space Grotesk

Web headings, release banners and architecture diagrams. Its geometric counters echo the round-stroke wordmark.

Body and UI `Text`

Inter

Long-form docs, API conventions, pack-format specs and Dear ImGui overlays (`F1` stats).

Code and C API `Mono`

JetBrains Mono

C99 signatures (`vgl_context_update`), CMake presets, CLI flags (`--depth reversed`) and shaders.

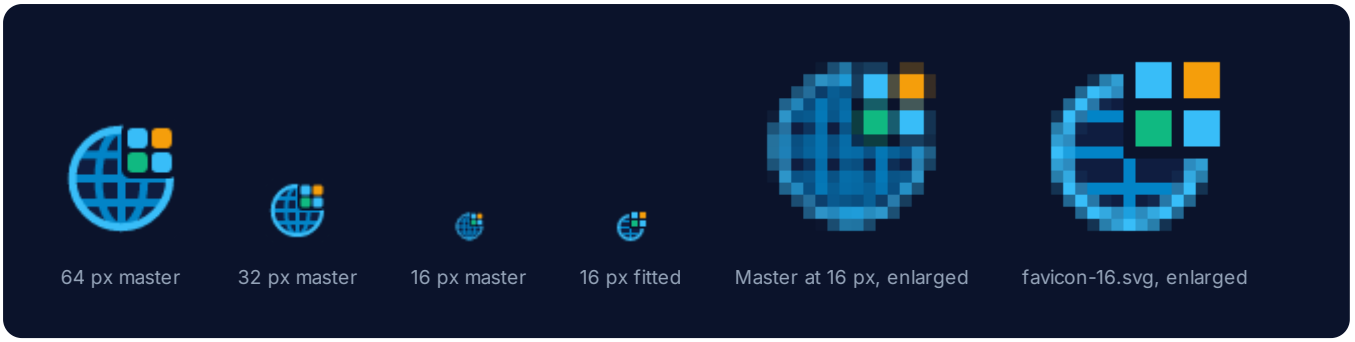
Role	Typeface	Weight	Case and tracking	Colour token
Wordmark (vgl)	Custom strokes, never typeset	11.5-unit stroke	Lowercase, fixed spacing	`--vgl-text` + dot
Lockup subtitle	Lato Bold, outlined to paths	700	Caps, tracked to the wordmark width	`--vgl-text-muted`
Headings	Space Grotesk	600 to 700	Sentence case, tracking -0.01em	`--vgl-text`
Body, tables, UI	Inter	400, 500, 600	Sentence case, default tracking	`--vgl-text`, `--vgl-text-muted`
Code, identifiers	JetBrains Mono	400, 600	As written in source	`--vgl-text`

Web type scale

Level	Size / line height	Face	Sample
H1	2.5 rem / 1.1	Space Grotesk 700	**Streaming the globe**
H2	1.75 rem / 1.2	Space Grotesk 600	**Building tile packs**
H3	1.25 rem / 1.3	Space Grotesk 600	**Reverse-Z depth**
Body	1 rem / 1.6	Inter 400	vgl streams terrain from local, range-readable tile packs.
Small	0.875 rem / 1.5	Inter 400	Last updated with vgl 0.1
Code	0.875 rem / 1.5	JetBrains Mono 400	`vgl_context_create(&desc, &ctx);`

Fallback stacks are in `css/vgl-tokens.css`. All four families are open source (SIL OFL) and available from Google Fonts.

07 Favicon and app icon scaling



The master holds up at 64 and 32 px, but at 16 px its 4-unit strokes fall to a single pixel and the graticule blurs. `favicon/favicon-16.svg` is a pixel-fitted redraw: 3×3 px tiles, a crisp meridian, equator and two parallels, and no meridian ellipse. `favicon.ico` uses it for the 16 px frame. Browsers that prefer `favicon.svg` render the master at 16 px; drop the SVG link if you want the fitted frame everywhere.

08 Master icon SVG

The equivalent of `svg/vgl-icon.svg` with its CSS classes written as attributes, so several copies can live in one HTML page without class collisions. It renders identically to the Illustrator master.

```
<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 64 64">
  <!-- tile -->
  <rect width="64" height="64" rx="14" fill="#0B132B"/>
  <!-- sphere fill and graticule -->
  <circle cx="31.91" cy="32.09" r="24.11" fill="#0F1D40"/>
  <g fill="none" stroke="#0284C7" stroke-width="4">
    <ellipse cx="31.91" cy="32.09" rx="11.35" ry="23.77"/>
    <path d="M31.91,8.31v47.54M8.14,32.09h47.54
             M11.38,20.2h41.06M11.38,43.97h41.06"/>
  </g>
  <!-- horizon ring -->
  <circle cx="31.81" cy="31.99" r="24.21" fill="none"
          stroke="#38BDF8" stroke-width="4"/>
  <!-- quadtree keyline and LOD tiles -->
  <rect x="32" y="4" width="28" height="28" rx="4.71" fill="#0B132B"/>
  <rect x="34.91" y="7"  width="10" height="10" rx="2.5" fill="#38BDF8"/>
  <rect x="46.91" y="7"  width="10" height="10" rx="2.5" fill="#F59E0B"/>
  <rect x="34.91" y="19" width="10" height="10" rx="2.5" fill="#10B981"/>
  <rect x="46.91" y="19" width="10" height="10" rx="2.5" fill="#38BDF8"/>
</svg>
```

09 Repository, web and UI integration

GitHub README header

Switches with the reader's GitHub theme. Paths assume the kit lives in `docs/assets/brand/`.

```
<p align="center">
  <picture>
    <source media="(prefers-color-scheme: dark)"
            srcset="docs/assets/brand/svg/vgl-logo-dark.svg">
    <img src="docs/assets/brand/svg/vgl-logo-light.svg"
         alt="vgl: Virtual Globe Library" width="360">
  </picture>
</p>
```

Website <head>

Copy the files from `favicon/` and `png/vgl-og-image.png` to the site root, and replace YOUR-SITE.

```
<link rel="icon" href="/favicon.ico" sizes="32x32">
<link rel="icon" href="/favicon.svg" type="image/svg+xml">
<link rel="apple-touch-icon" href="/apple-touch-icon.png">
<link rel="manifest" href="/site.webmanifest">
<meta name="theme-color" content="#0B132B">
<meta property="og:image" content="https://YOUR-SITE/vgl-og-image.png">
<meta property="og:image:width" content="1200">
<meta property="og:image:height" content="630">
<meta name="twitter:card" content="summary_large_image">
```



GitHub social preview

In the repository, open Settings, General, Social preview and upload `png/vgl-social-preview.png` (1280×640). Use `png/vgl-og-image.png` (1200×630) for the website's Open Graph image. Both are generated from the banner master on Orbital Slate with a 40 px construction grid.

10 Asset inventory

Everything ships in `vgl-brand-assets/`. After editing a master in Illustrator, run `python3 tools/build_brand_assets.py .` to regenerate the rest.

File	Contents	Derived from
`svg/vgl-banner.svg`	Master lockup on Orbital Slate	Master (Illustrator)
`svg/vgl-icon.svg`	64×64 app icon on tile	Master (Illustrator)
`svg/vgl-logo-dark.svg, -light.svg`	Horizontal lockup, transparent	Banner
`svg/vgl-logo-stacked-dark.svg, -light.svg`	Icon above wordmark, transparent	Banner
`svg/vgl-wordmark-dark.svg, -light.svg`	vgl letters only	Banner
`svg/vgl-mark.svg`	Globe mark, keyline knocked out	Icon
`svg/vgl-social-preview.svg / png`	GitHub social preview, 1280×640	Banner
`svg/vgl-og-image.svg / png`	Open Graph image, 1200×630	Banner
`png/*`	Lockups @1x/@2x, icon 64 to 1024 px, mark 256/512	Both
`favicon/favicon.ico`	16 (pixel-fitted), 32, 48 px frames	Icon
`favicon/favicon.svg, favicon-16.svg`	Master icon; pixel-fitted 16 px drawing	Icon
`favicon/apple-touch-icon.png, icon-*.png`	180, 192, 512, maskable 512	Icon
`favicon/site.webmanifest`	PWA manifest, theme colour Orbital Slate	n/a
`css/vgl-tokens.css`	Palette, type stacks, light/dark semantic tokens	n/a
`tools/build_brand_assets.py`	Regenerates every derived asset from the masters	n/a

11 Changes since v1.0

Area	Change
Icon geometry	Replaced with the Illustrator master: centred globe, 4-unit strokes, 2-unit tile gutter, 4.71 keyline radius. v1.0 listed an off-centre globe with 3 and 5-unit strokes.
Lockup and wordmark	Now defined by the vgl-banner.svg master. The g stem starts at the bowl's midline (no spur), and the icon placement is fixed by the master instead of re-derived.
Wordmark stroke	Unified at 11.5 units. The v1.0 brand kit said 8.5 while the manual said 11.5.
Contrast ratios	Recalculated with the WCAG 2.x formula. Bathymetry Blue is 4.49:1 (AA large / UI only, not AA body text). The kit's 14.2:1 for Deep Ocean was wrong (1.11:1). Ratios on white added.
Light mode	New: light-surface lockups and semantic tokens, with Slate 600 for muted text and Bathymetry Blue for the g dot.
Tokens	One naming scheme (--vgl-*) replaces the two conflicting sets in v1.0.
Typography	Roles reconciled: Lato Bold for the outlined subtitle only, Space Grotesk for display, Inter for body, JetBrains Mono for code.
Favicons	A pixel-fitted 16 px drawing replaces the claim that the master stays sharp at 16 px.
Rendering fixes	Asset filenames replace pasted image URLs; code blocks wrap instead of clipping; the truncated element table and orphaned section heading are gone; Starlight and Geoid Slate are split into separate rows.
Footer	Removed “Confidential & Proprietary”, which contradicted the MIT licence.